

K Nearest Neighbors

Type: Supervised Learning

Task: Classification and Regression

Steps Involved:

- Data Preparation: Organize labeled training data with features
- Choose K Value: Select number of neighbors to consider
- Distance Calculation: Compute distances between new point and all training points (Euclidean/Manhattan)
- Neighbor Identification: Find K closest instances
- Prediction:
 - Classification: Majority vote of neighbors' classes
 - Regression: Average of neighbors' values

Hyperparameters

- n_neighbors: Number of neighbors (K)
- weights: Distance weighting (uniform vs. weighted)
- algorithm: Computational method (brute, kd_tree, ball_tree)
- metric: Distance measurement (Euclidean, Manhattan)

Assumptions:

- Local data structure determines classification
- Features should be scaled/normalized
- No multicollinearity in features

Pros	Cons
Simple implementation	Computationally expensive with large data
No training phase	Sensitive to irrelevant features
Handles non-linear data	Requires feature scaling
No assumptions about data distribution	Poor performance with high dimensions

Example: Classifying a new customer as "high value" based on purchase patterns of their 5 nearest neighbors in historical data. Changing K from 3 to 11 might flip the prediction if border regions are ambiguous.